

In the footsteps of van der Waals — Solution

W22 (2022) — English translation

A1

0.40 pts

Write down an expression for finding the optimal value a_0 through the sums (or averages) of values T, V and their combinations.

We will solve the problem formally, express the temperature from the model function and write down the sum S that needs to be minimized.

$$T = \frac{(p + a_0) V}{R}$$

$$S = \sum_n \left(T_i - \frac{p V_i}{R} - \frac{a_0 V_i}{R} \right)^2$$

Differentiating:

$$\frac{\partial S}{\partial a_0} = \sum_n 2 \left(T_i - \frac{(p + a_0) V_i}{R} \right) \left(-\frac{V_i}{R} \right) = 0$$

Transforming (сократив на "-2"):

$$\frac{1}{R} \sum_n T_i V_i - \frac{p + a_0}{R^2} \sum_n V_i^2 = 0$$

And further

$$p + a_0 = \frac{R \sum_n T_i V_i}{\sum_n V_i^2}$$

We finally get

$$a_0 = \frac{R \sum_n T_i V_i}{\sum_n V_i^2} - p = R \frac{\overline{TV}}{\overline{V^2}} - p$$

Note: The answer could be written almost immediately if you realize that the model formula is linear dependence with one parameter: slope.

A2

0.30 pts

Calculate the sums (or averages) you need for the entire data set.

$\sum_n T_i V_i = 6.5903 \text{ K} \cdot ^3, \overline{TV} = 0.31382 \text{ K} \cdot ^3, \sum_n V_i^2 = 1.8935 \cdot 10^{-5} \text{ m}^6, \overline{V^2} = 9.0167 \cdot 10^{-7} \text{ m}^6$
Here and below, we save more significant figures to reduce the rounding error of the final answer.

A3

0.40 pts

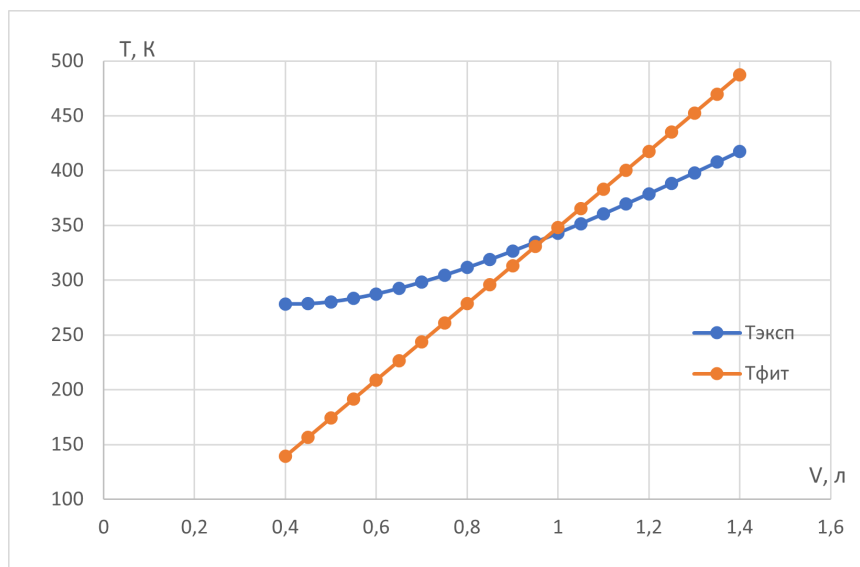
Calculate the value of a_0 . Specify it to three significant figures.

$$a_0 = 892 \text{ kPa}$$

A4

0.40 pts

For each volume value, calculate the temperature value that is obtained as a result of fitting. Plot the initial experimental data and fitting points on one graph. Mark data from different series with different icons. Sign the series.



B1

0.40 pts

Write down an expression for finding the optimal value a_1 through the sums (or averages) of values T, V and their combinations.

Again, we act formally:

$$T = \frac{pV}{R} + \frac{a_1}{R}$$

$$S = \sum_n \left(T_i - \frac{pV_i}{R} - \frac{a_1}{R} \right)^2$$

$$\frac{\partial S}{\partial a_1} = \sum_n 2 \left(T_i - \frac{pV_i}{R} - \frac{a_1}{R} \right) \left(-\frac{1}{R} \right) = 0$$

$$\sum_n T_i - \frac{p}{R} \sum_n V_i - \frac{a_1}{R} \sum_n 1 = 0$$

Hence we get окончательный answer:

$$a_1 = \frac{R \sum_n T_i - p \sum_n V_i}{n} = R\bar{T} - p\bar{V}$$

B2

0.30 pts

Calculate the sums (or averages) you need for the entire data set.

$$\sum_n T_i = 7009.2 \text{ K}, \bar{T} = 333.771 \text{ K}, \sum_n V_i = 0.0189 \text{ m}^3, \bar{V} = 0.0009 \text{ m}^3$$

B3

0.40 pts

Calculate the value of a_1 . Specify it to three significant figures.

$$a_1 = 974 \text{ Pa} \cdot \text{m}^3$$